

UART Design and Implementation using Verilog

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1. Introduction

- **UART (Universal Asynchronous Receiver Transmitter)** is a serial communication protocol which is used to exchange data between two devices.
- It is **asynchronous**, meaning the transmitter and receiver do not share a common clock, both operate on common parameter called as **baud rate** which tells receiver when to sample the data.
- Data is transmitted serially, between the **TX (transmit)** and **RX (receive)** lines.
- A standard UART frame generally consists of a **start bit, data bits, optional parity bit, and stop bit**.
- The Tx line remains at logic **HIGH when idle**, and it **pulled to low** for starting the transmission.
- **Oversampling** is often used in the receiver to improve the reliability of bit detection and tolerate small timing differences between the transmitter and receiver.

2. Formulas

UART protocol relies on Baud Rate, what we do is to identify after how many cycles we should transmit each bit and after how many cycles we should receive each bit based on respective clock frequencies of Transmitter, Receiver.

Cycles needed to transmit 1bit by Tx1 = $Tx1_Clk_Freq / BAUD_RATE$

Cycles needed to sample 1bit by Rx2 = $Rx2_Clk_Freq / (BAUD_RATE * OVERSAMPLE_FACTOR)$

1 bit of data transmitted by Tx is sampled multiple times using OVERSAMPLE_FACTOR by the receiver and only the middle sample is finally chosen.

3. Specifications

Parameter	Value
Baud Rate	115200 bps
Module 1 Frequency	50MHz

Module 2 Frequency	40MHz
Data Bits	8 bit
Parity	No
Start bit	Yes
Stop bit	Yes
Over Sampling Rate at Receiver	16x

4. Module Description

4.1 baud_rate_generator

Generates timing signals required to control the UART transmitter and receiver according to the selected baud rate.

4.2 Transmitter

Converts parallel data into a serial UART frame and transmits it bit-by-bit at the specified baud rate.

Inputs: Parallel data to be transmitted, a transmission-start request, and timing information from the baud-rate generator.

Outputs: Serial UART data and a status indication showing whether transmission is in progress.

4.3 Receiver

Detects and samples the incoming serial UART frame, reconstructs the transmitted byte, and indicates when reception is complete and outputs the parallel data.

Inputs: Incoming serial data, receiver enable, and sampling timing from the baud-rate generator.

Outputs: Reconstructed parallel data and a status indication showing that a complete byte has been received.

4.4 UART_Top_Module

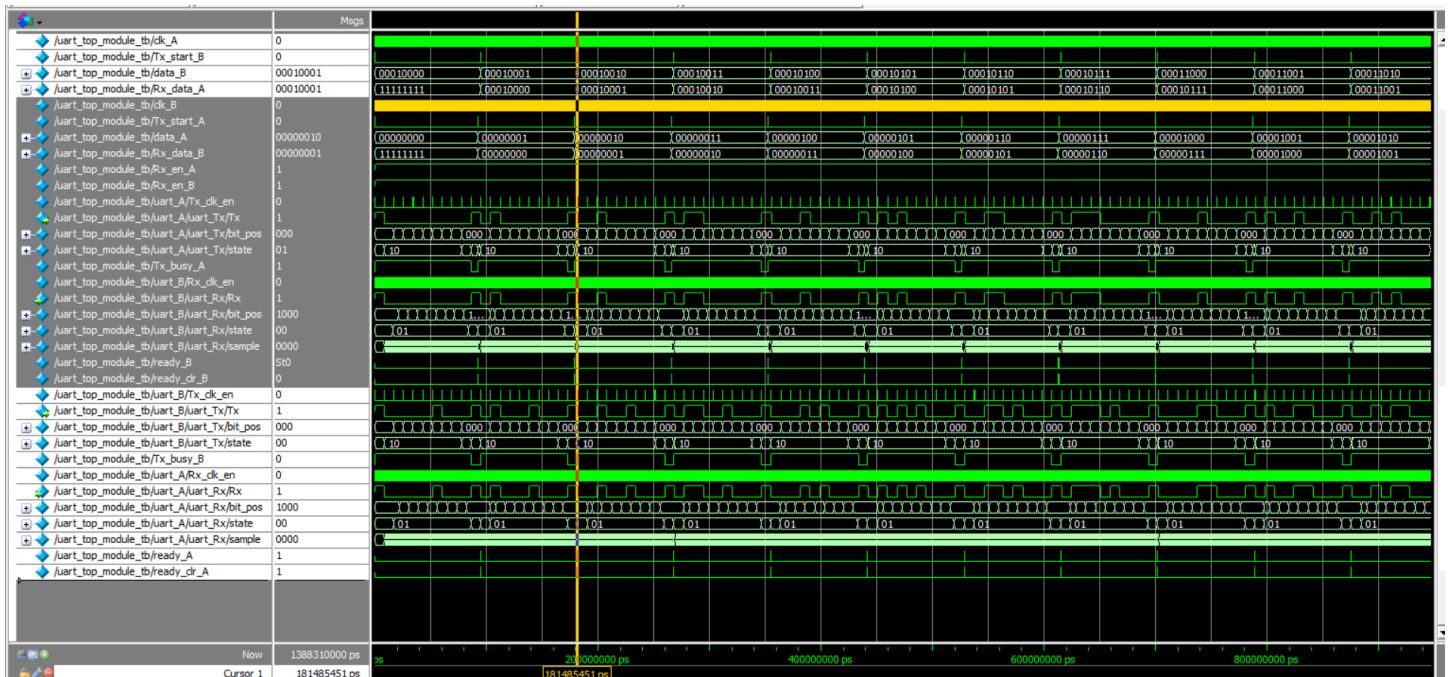
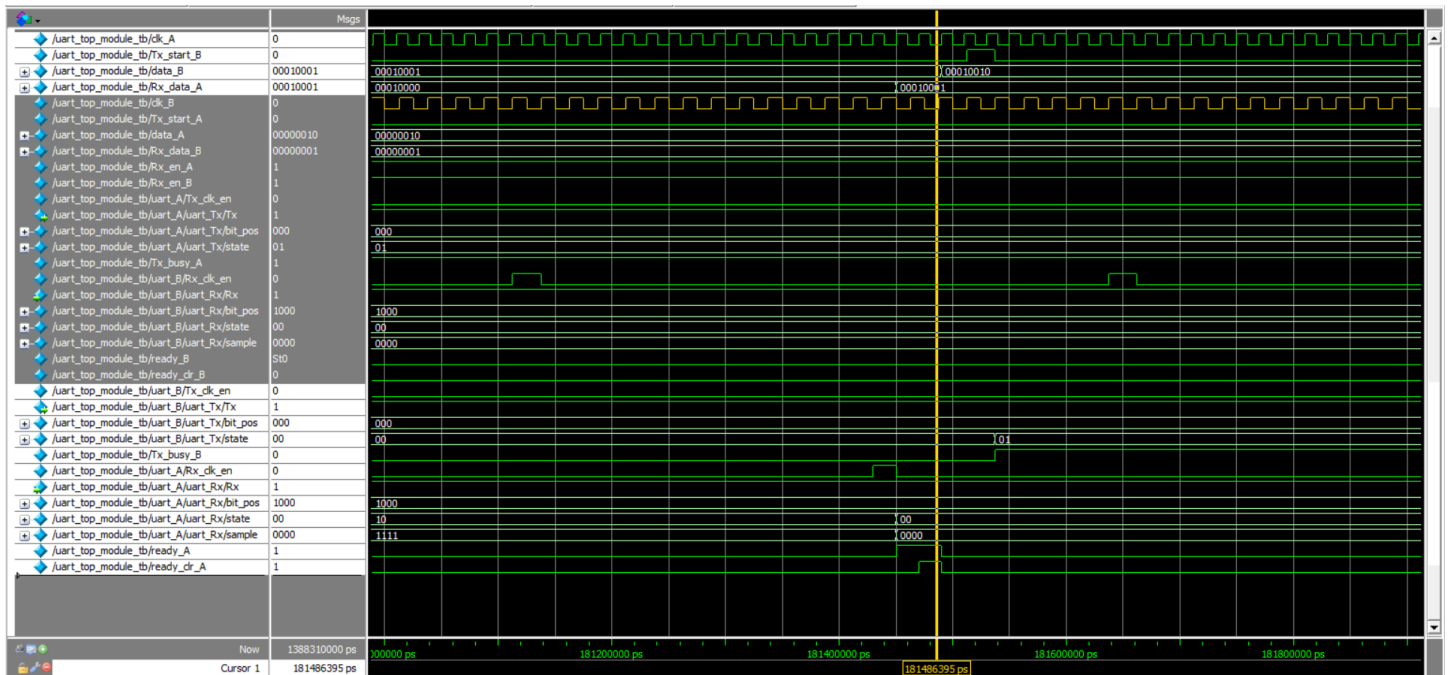
Integrates the baud-rate generator, transmitter, and receiver to form the complete UART system.

4.5 Testbench

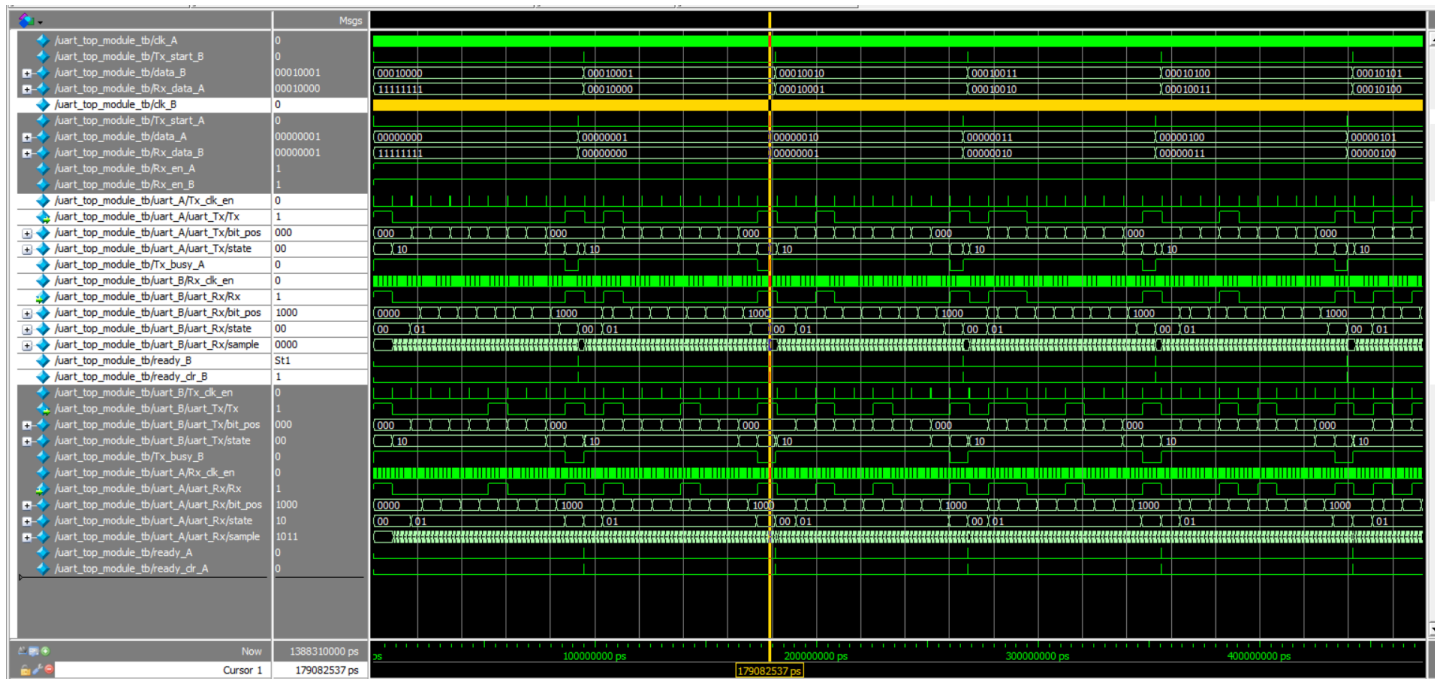
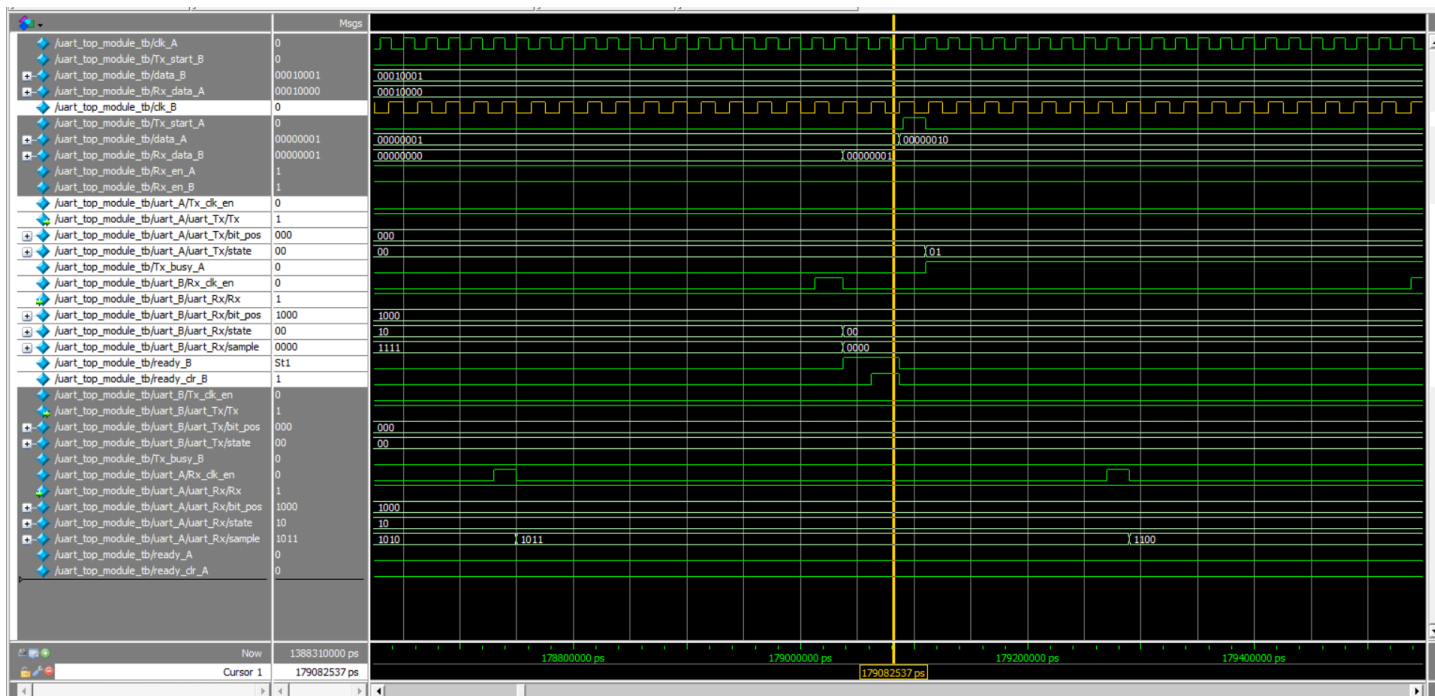
Verifies the functionality of the UART design by generating input data, clocks, and control signals and observing the transmitted and received data. It instantiates two UART modules

5. Waveforms

Data Sent from Tx of Module 2 to Rx of Module 1



Data Sent from Tx of Module 1 to Rx of Module 2



```
VSIM 69> run -all
```

```
# SUCCESS UART B: [time: 92463000] Rx data 00 match with data sent from Tx of moduleA 00  
# SUCCESS UART A: [time: 95090000] Rx data 10 match with data sent from Tx of moduleB 10  
# SUCCESS UART B: [time: 179088000] Rx data 01 match with data sent from Tx of moduleA 01  
# SUCCESS UART A: [time: 181490000] Rx data 11 match with data sent from Tx of moduleB 11  
# SUCCESS UART B: [time: 266238000] Rx data 02 match with data sent from Tx of moduleA 02  
# SUCCESS UART A: [time: 268430000] Rx data 12 match with data sent from Tx of moduleB 12  
# SUCCESS UART B: [time: 352863000] Rx data 03 match with data sent from Tx of moduleA 03  
# SUCCESS UART A: [time: 355370000] Rx data 13 match with data sent from Tx of moduleB 13  
# SUCCESS UART B: [time: 439488000] Rx data 04 match with data sent from Tx of moduleA 04  
# SUCCESS UART A: [time: 441770000] Rx data 14 match with data sent from Tx of moduleB 14  
# SUCCESS UART B: [time: 526638000] Rx data 05 match with data sent from Tx of moduleA 05  
# SUCCESS UART A: [time: 528710000] Rx data 15 match with data sent from Tx of moduleB 15  
# SUCCESS UART B: [time: 613263000] Rx data 06 match with data sent from Tx of moduleA 06
```

```
Now: 1,388,310 ns Delta: 3
```

```
St1
```

6. Conclusion

Successfully Implemented UART showing full duplex data transfer with serial data transmission.